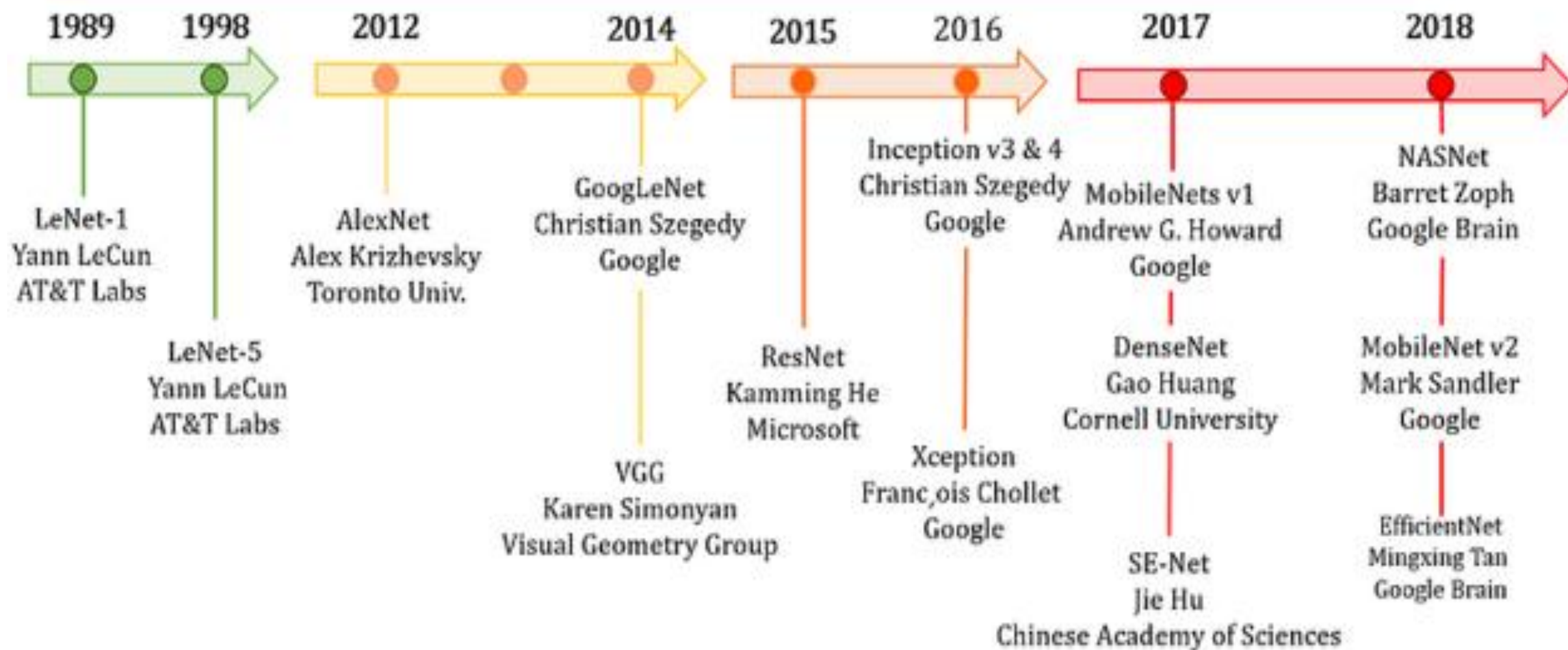


Pre-trained Models

CNN Timeline (1989 ~ 2018)

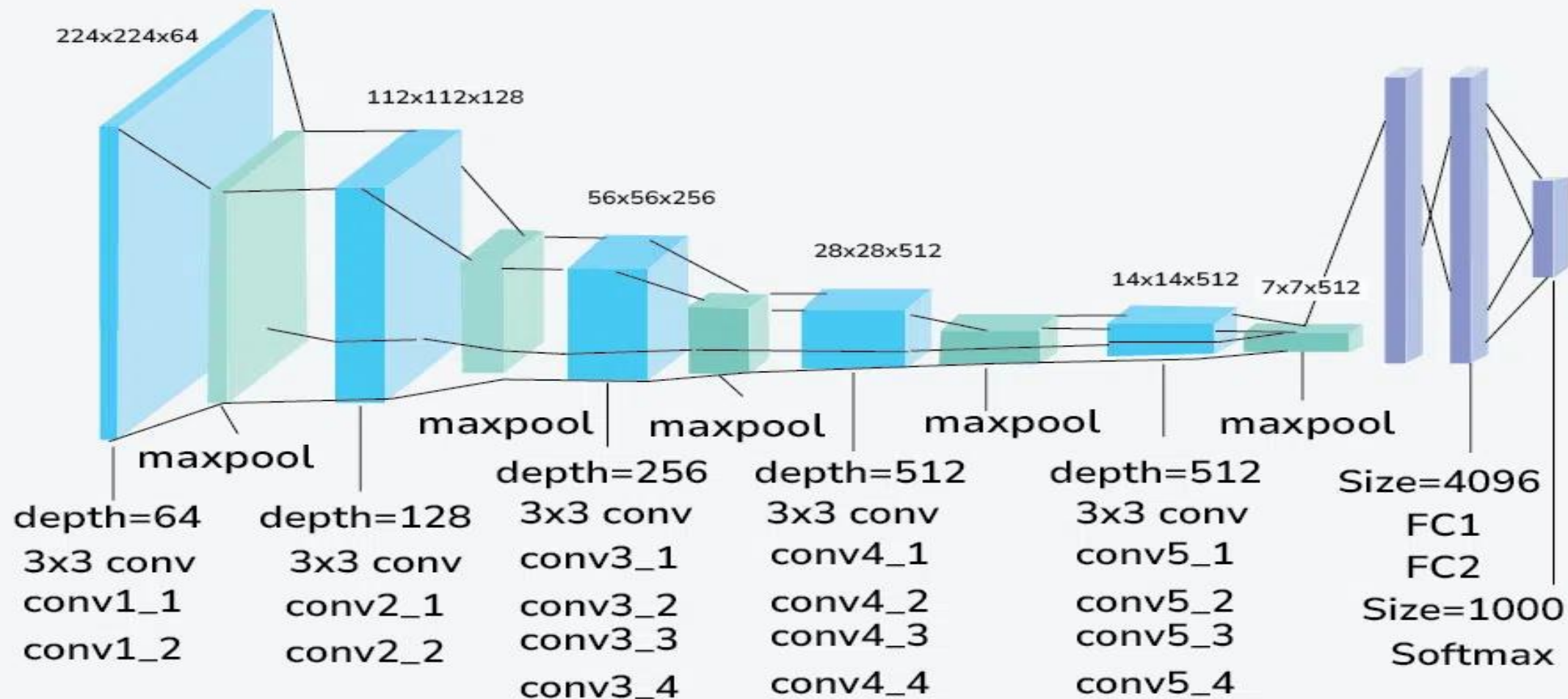


CNN	Year	Key Objective	Special Features / Innovation
LeNet-5	1998	Handwritten digit recognition	Early CNN; convolution + pooling + fully connected
AlexNet	2012	Image classification on ImageNet	ReLU, dropout, data augmentation; deeper than LeNet
VGGNet	2014	Accurate image classification	Very deep (16–19 layers), small 3×3 filters; simple and uniform
GoogLeNet / Inception	2014	Efficient deep classification	Inception modules, parallel convolutions, global average pooling
ResNet	2015	Enable very deep networks	Residual (skip) connections; solves vanishing gradient
DenseNet	2016	Efficient deep networks	Dense connectivity (all layers connected); feature reuse
MobileNet	2017	Lightweight models for mobile	Depthwise separable convolutions; low memory footprint
EfficientNet	2019	High accuracy with low computation	Compound scaling (depth, width, resolution); state-of-the-art

Visual Geometry Group (VGG) models

- The ***Visual Geometry Group (VGG) models***, particularly ***VGG-16 and VGG-19***, have significantly influenced the field of computer vision since their inception. These models, introduced by the Visual Geometry Group from the University of Oxford, stood out in the 2014 ImageNet Large Scale Visual Recognition Challenge (ILSVRC) for their deep convolutional neural networks (CNNs) with a uniform architecture. VGG-19, the deeper variant of the VGG models, has garnered considerable attention due to its simplicity and effectiveness.

VGG -19 Architecture



VGG 19

Block	Layers	Purpose	Where Input Goes
Input Layer	Input tensor (224,224,3)	Receives RGB image	✓ First step, outside all convolution blocks
Block 1	Conv(64) → Conv(64) → MaxPool	Low-level feature extraction (edges, textures)	Receives the image directly
Block 2	Conv(128) → Conv(128) → MaxPool	Mid-level features (shapes, corners)	Output of Block 1
Block 3	Conv(256) × 4 → MaxPool	More complex features	Output of Block 2
Block 4	Conv(512) × 4 → MaxPool	High-level features	Output of Block 3
Block 5	Conv(512) × 4 → MaxPool	Very abstract features	Output of Block 4
Flatten + Custom Dense Layers	Dense(512) → Dropout → Dense(num_classes)	Classification	Output of Block 5

Code

```
import tensorflow as tf  
from tensorflow.keras.applications import VGG19  
from tensorflow.keras.models import Model
```

```
base_model = VGG19(weights='imagenet', include_top=False,  
input_shape=(224, 224, 3))
```